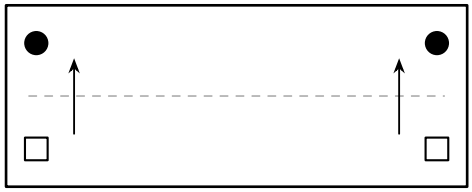
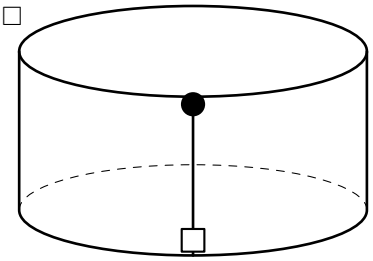
Keep the bands on the table. An adult makes every cut. The rectangles are joining recipes.

input process output



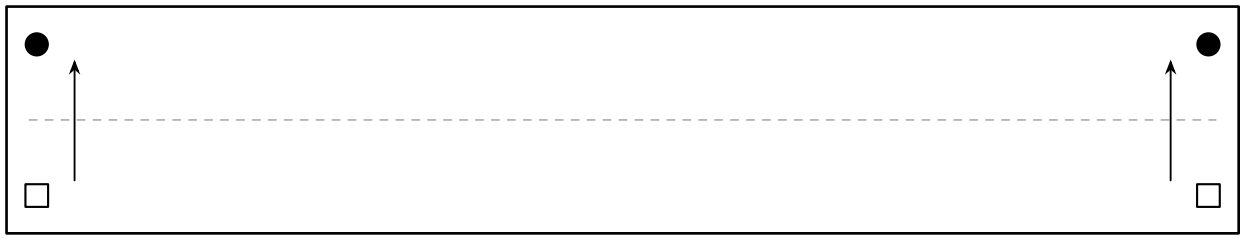
Join • to • and □ to □.



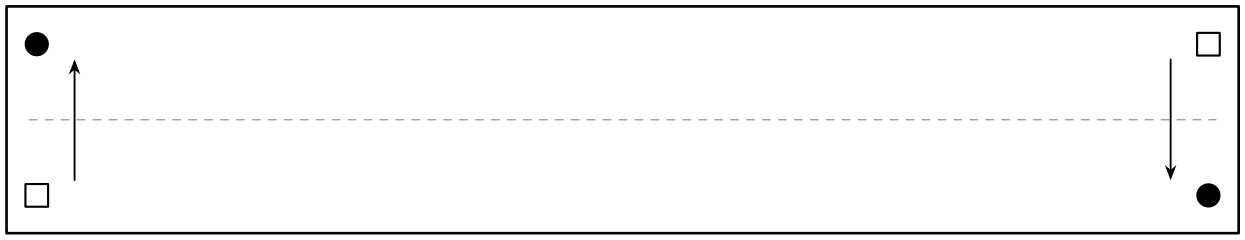
An edge trip stays on an edge, follows its seam marks, and stops on its first return to the start dot.

Problem 1: Trace all the edges of A and B. Show on each flat recipe where each separate closed edge goes through the seam.

A



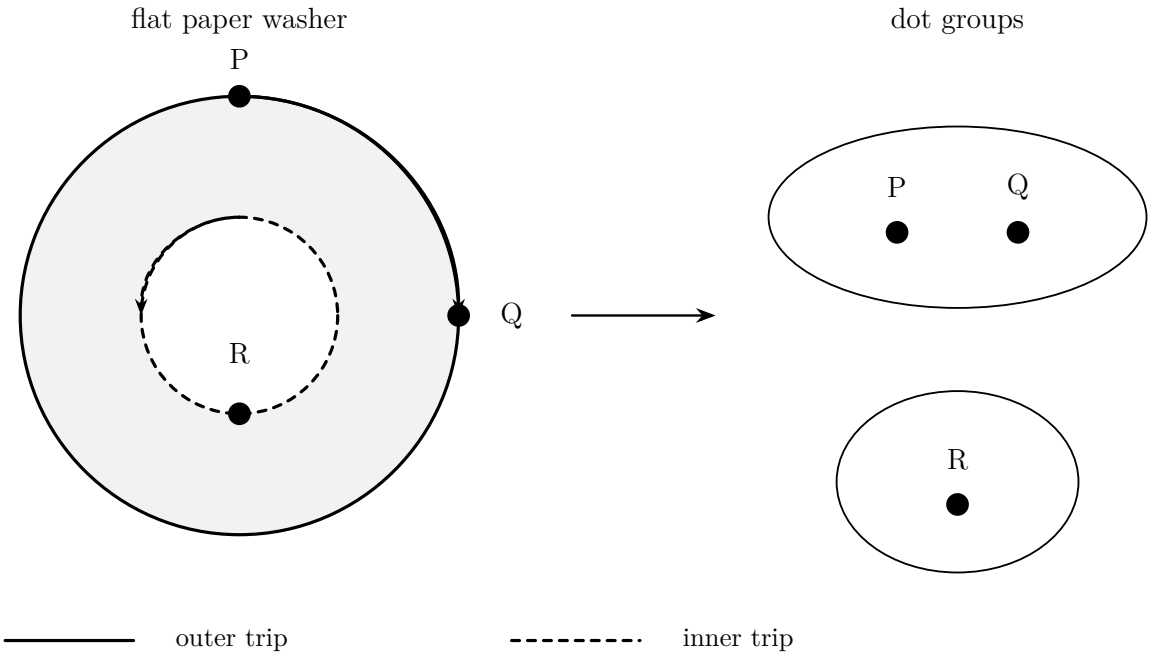
B



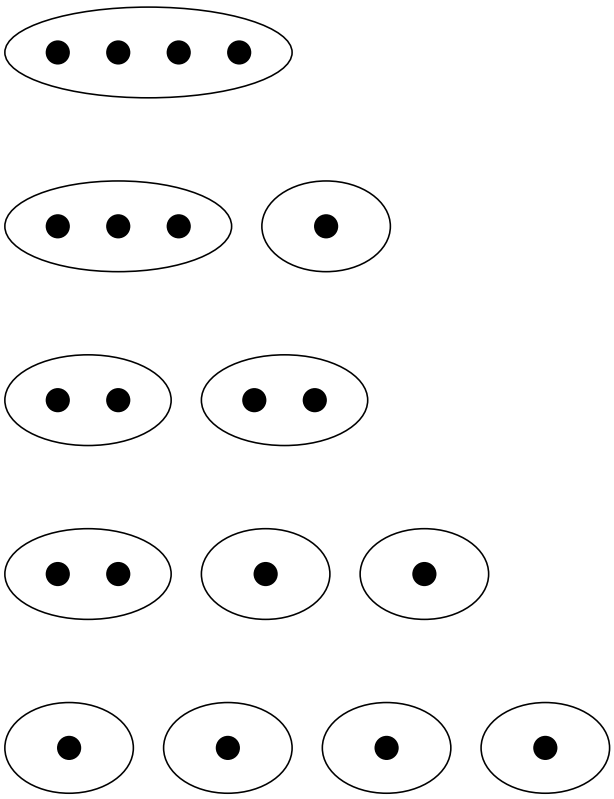
A: _____ B: _____

Problem 2: One separate closed edge is a boundary component. Explain the number of boundary components of A and B from their seam recipes.

Dots on one edge go in one ring.



Problem 3: Place four edge dots to match each picture on A and on B. Write yes or no in each blank. Each ring means a different closed edge; other edges may have no dots.



A: _____ B: _____

A: _____ B: _____

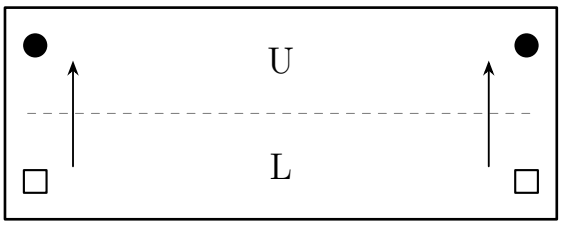
A: _____ B: _____

A: _____ B: _____

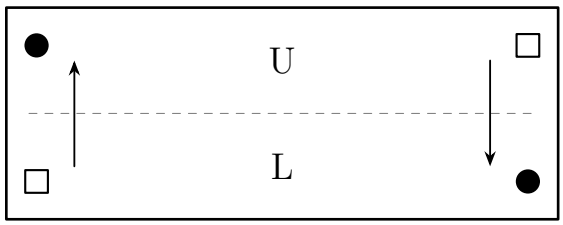
A: _____ B: _____

Problem 4: Predict the number of connected pieces after a middle-line cut in A and B. Have an adult cut both bands, then count the pieces and the boundary components of each piece.

A



B



I predict: _____

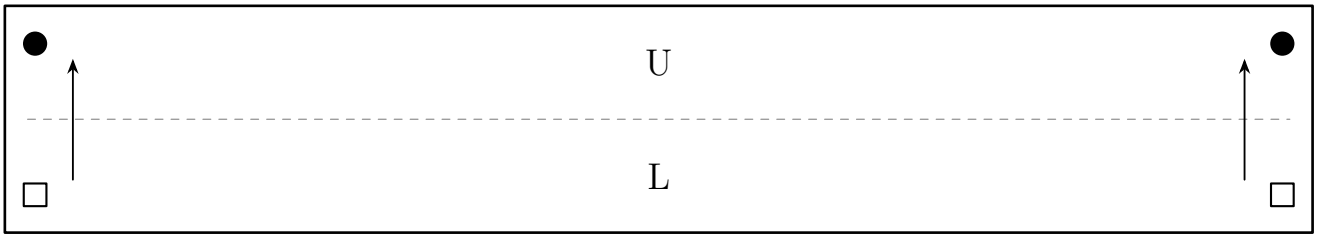
I predict: _____

after cutting A

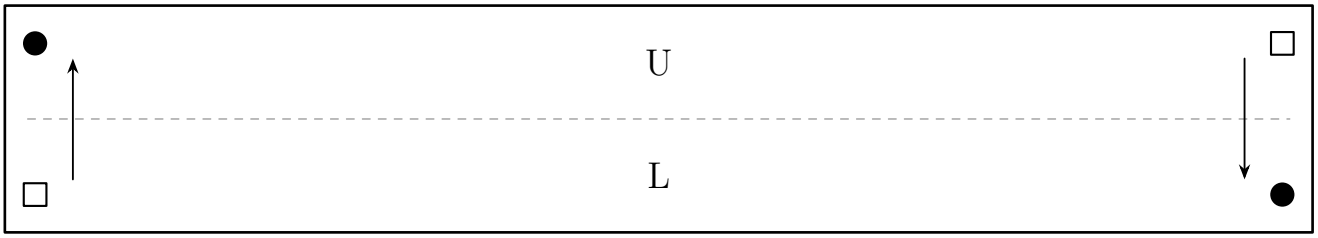
after cutting B

Problem 5: Use U and L to explain the connected pieces after the cuts. Explain separately the number of boundary components on each piece.

A



B



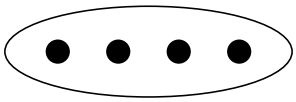
Problem 6: Ask an adult to rebuild uncut A and B from the two spare strips. Put a removable arrow across each strip, pointing from U toward L. Slide it along the middle line for one full trip, keeping it across the strip.



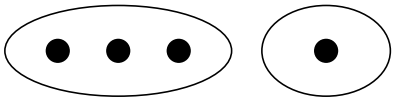
start marker after a short slide

Compare the returning arrow with its starting position. Explain its return direction from the seam.

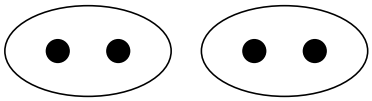
Problem 7: Which four-dot pictures from Problem 3 are possible after each cut? Write yes or no in each blank, and explain.



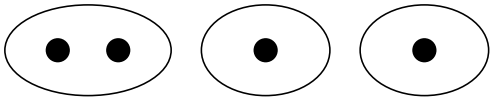
cut A: _____ cut B: _____



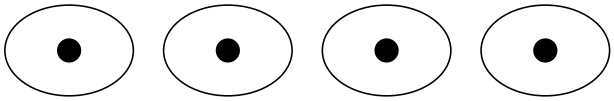
cut A: _____ cut B: _____



cut A: _____ cut B: _____



cut A: _____ cut B: _____



cut A: _____ cut B: _____